

CASE 4

COMMON VARIABLE IMMUNODEFICIENCY

A failure to produce antibodies against particular antigens.

The ability to produce antibody of all different classes after exposure to antigen is an important aspect of successful and comprehensive humoral immunity. Although antibody production and class switching in response to most protein antigens require helper T cells (see Case 2), responses to some other antigens do not. This is because the special properties of certain bacterial polysaccharides, polymeric proteins, and lipopolysaccharides enable them to stimulate naive B cells in the absence of T-cell help. These antigens are known as thymus-independent antigens (TI antigens) because they stimulate strong antibody responses in athymic individuals. They fall into two classes: TI-1 antigens, such as bacterial lipopolysaccharide, which can directly induce B-cell division; and TI-2 antigens, such as bacterial capsular polysaccharides, which do not have this property but have highly repetitive structures and probably stimulate the B cell by cross-linking a critical number of B-cell receptors. In particular, B-cell responses to thymus-independent antigens provide a prompt and specific IgM response to an important class of pathogen-encapsulated bacteria. Pyogenic bacteria such as streptococci and staphylococci are surrounded by a polysaccharide capsule that enables them to resist phagocytosis. Antibody that is produced rapidly in response to this polysaccharide capsule can coat these bacteria, promoting their ingestion and destruction by phagocytes.

Whereas TI-1 antigens are inefficient inducers of affinity maturation and memory B cells, TI-2 antigens can induce both IgM and some class-switched responses. The initiation of B-cell class switching usually depends on the interaction of B cells and helper T cells via CD40 and CD40 ligand (see Case 2). Class switching in response to TI antigens is thought to involve the two members of the tumor necrosis factor (TNF)/TNF receptor (TNFR) family BAFF (B-cell activating factor belonging to the TNF family) and APRIL (a proliferation-inducing TNF ligand), and their receptors on B cells. In human B cells, BAFF and APRIL induce class switching to IgA and IgG in the presence of TGF- β or IL-10 and to IgE in the presence of IL-4. In mice, in contrast, BAFF and APRIL on their own can switch B cells to IgG and IgA production.

BAFF is secreted by cells in the follicles of peripheral lymphoid tissue and can bind to several receptor proteins on B cells, which include BAFF receptor (BAFF-R), transmembrane activator and calcium-modulating cyclophilin ligand interactor

TOPICS BEARING ON THIS CASE:

Antibody classes

Class switching

Thymus-independent
antigens

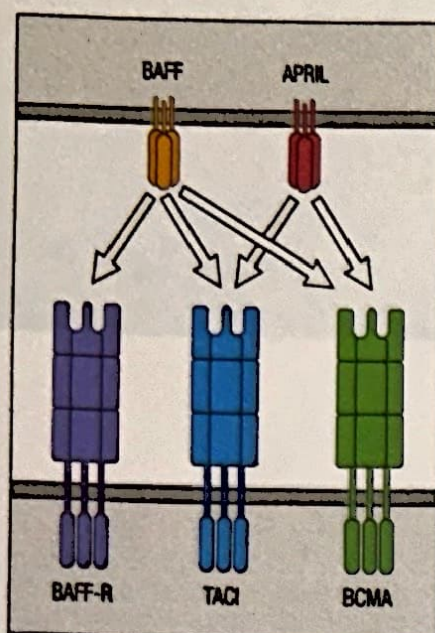


Fig. 4.1 BAFF and APRIL, and their receptors on B cells. The BAFF receptor (BAFF-R) is expressed on resting B cells and mediates the effects of BAFF on B-cell development and survival but has only a minor role in isotype switching. Studies in mice suggest that interaction of BAFF with BAFF-R has an important role in B-cell survival and is required for the development of mature B cells *in vivo*. TACI seems to be the receptor that mainly mediates isotype switching by BAFF and APRIL. TACI may also have other functions such as promoting plasma-cell differentiation and survival. BCMA is poorly expressed on resting B cells but is upregulated on germinal-center B cells and plasma cells. It has the ability to upregulate the expression of co-stimulatory molecules on the B-cell surface that enhance antigen presentation and T-cell activation, and has a role in maintaining the survival of long-lived plasma cells. BCMA has no detectable role in isotype switching by BAFF and APRIL.

(TACI), and B-cell maturation antigen (BCMA) (Fig. 4.1). In addition to its role in class switching, BAFF is involved in B-cell development and in promoting the survival of mature B cells. Mice carrying a BAFF transgene, leading to BAFF overexpression, develop high titers of autoantibodies and a systemic lupus erythematosus-like condition, whereas BAFF-deficient mice have a severe defect in B-cell development and an almost complete loss of mature B cells and of follicular and marginal zone B cells in lymph nodes and spleen. A similar phenotype is observed in mice lacking BAFF-R, or in the A/WySnJ mouse strain, which carries a naturally occurring mutation in BAFF-R.

APRIL is the closest relative of BAFF in the TNF family, sharing 33% sequence identity, and was initially reported as a protein that stimulated the proliferation of tumor cells. Unlike BAFF, it does not seem to have an effect on B-cell development and survival, because APRIL-deficient mice have normal numbers of B cells in all developmental stages. This is because APRIL binds to TACI and BCMA, but not to BAFF-R. The only detectable immune defects in APRIL homozygous null mice are diminished serum IgA levels and an impaired antibody response to immunization by the oral route. TACI^{+/-} mice exhibit a marked haploinsufficiency in regard to their ability to mount an antibody response to type II dependent antigens *in vivo* and to the ability of their B cells to respond to APRIL *in vitro*.

The receptors BAFF-R, TACI, and BCMA are all members of the TNFR superfamily and have different roles in immune responses. TACI seems to be the receptor that mainly mediates isotype switching by BAFF and APRIL. BCMA, and possibly TACI, may promote plasma-cell differentiation and survival. TACI-deficient mice have low levels of serum IgA and have a deficient antibody response to immunization with TI-2 antigens in the vaccine Pneumovax, which contains polysaccharide antigens from a number of *Streptococcus pneumoniae* serotypes, and NP-Ficoll, a polysaccharide. They also have enlarged spleens and lymph nodes, with more cells than usual and an increased number of B cells. They develop autoimmunity, suggesting that TACI may have a regulatory role in B cells. Some cases of common variable immunodeficiency (CVID) in humans have recently been shown to be associated with mutations in TACI, as we shall see in this case study.

The case of Mary Johnson: impaired ability to generate all classes of antibodies leads to frequent and unusual infections.

Mrs Johnson was 40 years old when she was referred to the Immunology clinic for evaluation of her immune system after suffering throughout her life from recurrent respiratory and gastrointestinal infections. As a child she was frequently diagnosed with otitis, sinusitis, and tonsillitis and had intermittent diarrhea, and from an early age she was hospitalized several times for pneumonia and gastrointestinal infections. The year before she came to the Immunology clinic she had been in hospital with severe diarrhea caused by an infection with the protozoan parasite *Giardia lamblia*. When she was 25 years old, Mrs Johnson had been diagnosed with thyroid insufficiency and placed on thyroid hormone replacement therapy.

Physical examination revealed an enlarged spleen, the edge of which extended 8 cm below the left mid-costal margin. Blood tests showed that Mrs Johnson had lower than normal levels of all immunoglobulin isotypes: IgM 18 mg dl⁻¹ (normal 100–200 mg dl⁻¹), IgG 260 mg dl⁻¹ (normal 600–1000 mg dl⁻¹), and IgA 24 mg dl⁻¹ (normal 60–200 mg dl⁻¹). Although she had been immunized several times with pneumococcal vaccines, she had been unable to respond, as shown by her lack of antibodies against all pneumococcal serotypes present in the vaccine. Numbers of B cells and T cells were normal. No antinuclear autoantibodies or rheumatoid factor

40-year-old woman.
Recurrent infections.

(an autoantibody against IgG) were detected, but Mrs Johnson had high levels of antithyroid antibodies.

Mrs Johnson's twin sister and her mother were both dead. They had also had hypogammaglobulinemia and an inability to make specific antibodies against polysaccharide vaccines. From her early teens onward, the sister had had recurrent viral and bacterial infections. She developed hemolytic anemia and granulomatous vasculitis, and died at the age of 31 years from gastrointestinal cancer. The mother had died of non-Hodgkin's lymphoma. Mrs Johnson's only brother had been diagnosed with common variable immune deficiency (CVID) and suffered from chronic sinusitis and recurrent chest infections. Her father was still alive.

On the basis of the blood tests and her family history, Mrs Johnson was diagnosed with CVID and was placed on intravenous immunoglobulin 35 g every 2 weeks. This caused a remarkable improvement in her condition with a marked decrease in the frequency of infections. Sequencing of the *TACI* gene revealed that Mrs Johnson had a mutation in one of the alleles. The same mutation was also found in her brother's *TACI* gene but was absent from their father's DNA, implying that she and her brother had inherited the mutation from their mother.

Family history of
CVID.

Start immunoglobulin
therapy.

Common variable immunodeficiency (CVID).

CVID is an immunodeficiency disorder characterized by low serum levels of all switched immunoglobulin isotypes (IgG, IgA, and IgE), an impaired ability to produce specific antibodies, even of the IgM class, after exposure to certain antigens, and increased susceptibility to infections of the respiratory and gastrointestinal tract, the latter being due to the combined decrease in IgG and IgA. It is relatively common (hence the name), being the most common primary immunodeficiency that comes to medical attention. Patients with CVID have a striking paucity of both unswitched IgM⁺ CD27⁺ memory B cells and of switched IgM⁺ CD27⁺ memory B cells (Fig. 4.2). In addition they are severely deficient in plasma cells and have impaired somatic hypermutation. Consequently, they are hypogammaglobulinemic and the few antibodies they develop have diminished affinity, rendering them susceptible to chronic and recurrent infections by encapsulated bacteria such as *S. pneumoniae* (pneumococcus) or *Haemophilus influenzae*. The clinical course, as well as the degree of deficiency of serum immunoglobulins, varies from patient to patient (hence the name 'variable').

Like other immunodeficiency diseases, the symptoms of CVID are frequent and unusual infections. Symptoms may appear during early childhood, adolescence, or adult life, but the age of onset of symptoms is usually in the twenties or thirties

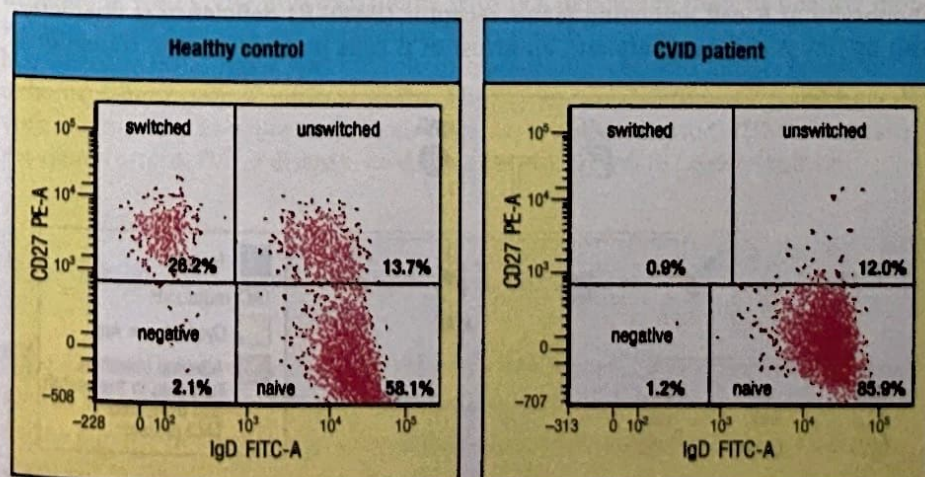


Fig. 4.2 CVID is associated with reduced numbers of switched and unswitched memory B cells. Flow-cytometric measurement of the memory B cell compartment is shown for a healthy control (left) and a patient with CVID (right). Blood lymphocytes are gated on B cells (CD19). The upper panels show CD27⁺ memory B cells. In the left panels are B cells that underwent class switching (IgD⁺), and in the right panels are unswitched B cells (IgD⁺) that still carry IgD on their surface. Measurement of switched and unswitched memory B cells has become part of the routine evaluation of patients with CVID. Low percentages of switched memory B cells (<2%) predict disease-associated clinical complications such as splenomegaly and autoimmune disease.

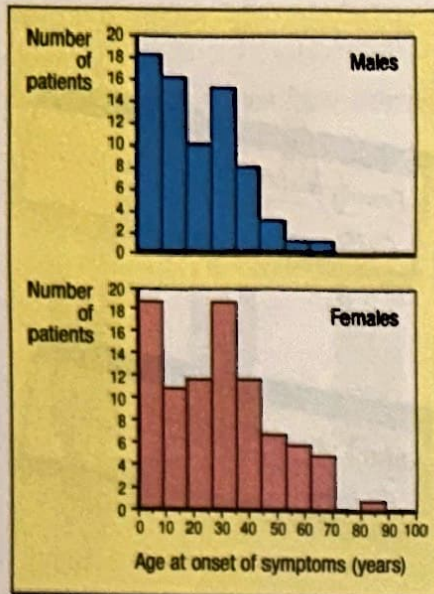


Fig. 4.3 The age at onset of clinical symptoms of immunodeficiency in CVID. The age at onset of clinical symptoms of immunodeficiency is shown in centiles. The median age at onset of symptoms is 23 years for males and 28 years for females.

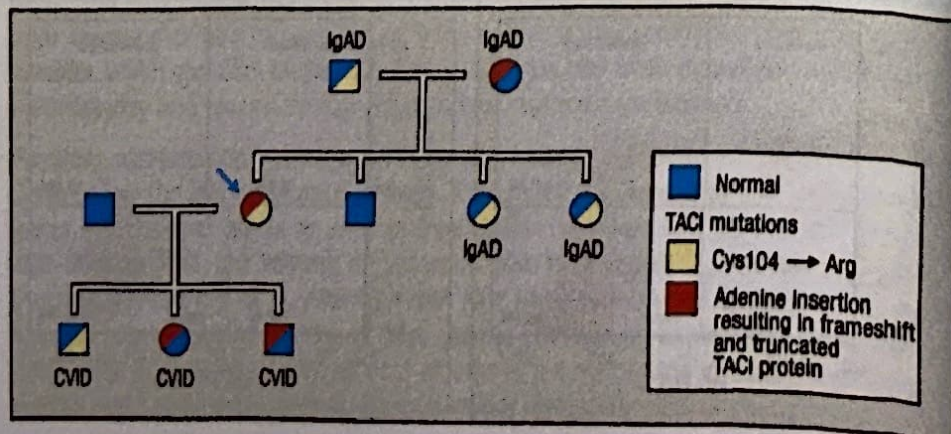
Fig. 4.4 Illustrative pedigree of a family with IgA deficiency (IgAD) and CVID. The index CVID case is marked with an arrow. Note the presence of both CVID and IgAD in family members with the same TACI genotype. Two different mutations in TACI have been found in this pedigree: C104R, which destroys ligand binding; and an insertion, A204bp, which results in a frameshift and premature termination of translation. Squares indicate males, circles females. Fully shaded symbols indicate homozygosity, half-shaded symbols heterozygosity. Most heterozygous TACI mutations, including C104R, cause functional haploinsufficiency with regard to the response to T-independent type II antigens (for example pneumococcal antigens). They are disease modifiers, because they are found in 10% of CVID patients but also in 1% of normal individuals.

(Fig. 4.3). Patients with CVID suffer from recurrent infections of the sinuses and lungs, the ears, and the gastrointestinal tract. If therapy is delayed, the air passages in the lungs may become irreversibly damaged and chronic infections may develop (bronchiectasis). Patients with CVID also are at greater risk of developing autoimmune diseases such as autoimmune thyroiditis, hemolytic anemia, autoimmune thrombocytopenia (which is due to antiplatelet antibodies), and pernicious anemia (which is caused by antibodies against the intrinsic factor that is required for the absorption of vitamin B12). Patients with CVID also have a 300-fold increased risk of lymphoma and a 50-fold increased risk of gastric carcinoma. Some patients with CVID develop granulomatous lesions (see Case 26) in the lungs and the skin that are characterized by the presence of T cells and macrophages. Human herpes virus 6 (HHV6) has been isolated from these lesions, which usually respond to immunosuppressive treatment with corticosteroids or cyclosporin.

Most cases of CVID are sporadic; that is, they are not due to an inherited defect. Studies in European populations show that around 20% of CVID cases are familial. Genetic analysis has shown a high degree of familial clustering of single IgA deficiency (a deficiency of IgA antibodies only, which usually does not produce any detectable symptoms) and CVID, suggesting that defects in the same genes might underlie both diseases (Fig. 4.4). Moreover, patients may first present with IgA deficiency and, after several years, may develop a full-blown picture of CVID.

The genetic and immunologic causes of CVID have been largely unknown. The heterogeneous nature of the disease is demonstrated by documentation of defects in T cells, B cells, and antigen-presenting cells, suggesting that many genes are involved. It has, however, recently been demonstrated that some familial CVID cases might be monogenic disorders. In the past few years, mutations in several genes have been described that either cause or contribute to CVID. Mutations in the gene encoding TACI have been identified in 8–10% of patients with CVID. Both homozygosity and heterozygosity for mutations in TACI have been shown to be associated with familial and sporadic cases of CVID (see Fig. 4.4). However, mutations in TACI have been detected in 1% of individuals who do not suffer recurrent infections, including family members of index cases. This reflects the variable penetrance of the gene defect, and suggests that the TACI mutation might synergize with defects in other genes to cause CVID. In this regard the majority of patients with CVID have poor B-cell responses to stimulation of the Toll-like receptor TLR-9, and data from mice suggest that TACI synergizes with TLR-9 and also with CD40 in promoting plasma-cell differentiation and immunoglobulin secretion. Naive B cells from CVID patients with TACI mutations are severely impaired in their ability to secrete IgG and IgA in response to stimulation with APRIL *in vitro*. In addition, patients with TACI mutations demonstrate an impaired antibody response to the pneumococcal vaccine Pneumovax.

Other gene defects have been identified as very rare causes of CVID (Fig. 4.5). The phenotype of B-cell activating factor receptor (BAFF-R) deficiency is characterized by low B-cell numbers and an arrest of B cells in an immature transitional



stage, which can be explained by the critical role of BAFF-R in promoting B-cell development and survival. The few patients described so far developed late-onset hypogammaglobulinemia with recurrent infections.

A number of B-cell intrinsic defects that result in CVID affect signaling through the B cell co-receptor CD19. CD19 is a B-cell surface protein that cooperates in a complex with the complement receptor CD21 and with the surface molecule CD81 in lowering the B-cell-receptor signaling threshold after antigen binding. Described patients with CD19 deficiency presented in childhood with recurrent infections and hypogammaglobulinemia. Because CD19 is a surface marker for B cells, its expression is routinely assessed by flow cytometry, when lymphocyte subsets are analyzed. CD21 deficiency has also been identified as a cause for CVID and is associated with a milder clinical phenotype and impairment of the antibody response to pneumococcal polysaccharides. A mutation in the *CD81* gene has also been shown to affect the B-cell co-receptor complex, resulting in CVID. A reported patient with a homozygous *CD81* mutation had normal *CD19* alleles, but no membrane expression of CD81 and CD19 molecules on B cells, with an associated disruption of the CD19 signaling complex. The patient's immunologic phenotype was characterized by impaired specific antibody responses and a decreased number of memory B cells.

A homozygous mutation in the gene encoding CD20, another B-cell surface marker, has been described as causing CVID in one patient. CD20 has a role in the regulation of Ca^{2+} transport across the plasma membrane. CD20 deficiency in this patient was associated with decreased IgG serum levels, a severe reduction of class-switched memory B cells, and an inability to mount T-independent antibody responses.

Deficiency of the inducible co-stimulator (ICOS) molecule is another very rare cause of CVID. ICOS is a co-stimulatory molecule that is upregulated on activated T cells and has a critical role in T-cell-B-cell cooperation. Patients have hypogammaglobulinemia, reduced B-cell numbers, and markedly reduced numbers of memory B cells. All identified patients have the same homozygous mutation in ICOS and are believed to descend from a common ancestor.

The most recently discovered single-gene defect that causes CVID is a mutation in LPS-responsive beige-like anchor (LRBA). LRBA is a cytosolic protein thought to be involved in intracellular vesicle trafficking and internalization of ligand-activated receptors, with its entire function still not being completely understood (see Case 53). Patients with a homozygous truncated *LRBA* mutation have abolished protein expression, which leads to a clinical presentation characterized by immunodeficiency and immune dysregulation with prominent inflammatory bowel disease and autoimmune manifestations.

Lastly, several single nucleotide polymorphisms (SNPs) in the gene encoding the mismatch repair (MMR) protein Msh5 have been found at greater frequency in patients with CVID and in patients with IgA deficiency than in healthy individuals. MMR proteins are involved in isotype switching, as discussed in Case 12, but whether Msh5 has a specific role in class switch recombination is still being debated. These polymorphisms in the *Msh5* gene are also found in healthy individuals; they are therefore more likely to cause susceptibility to CVID than to cause the disease itself. Other disease-modifying factors are yet to be determined.

Genetic defects in CVID	Relative frequency
TACI	very common
BAFF-R	extremely rare
CD19	extremely rare
CD20	extremely rare
ICOS	extremely rare
CD21	extremely rare
CD81	extremely rare
LRBA	extremely rare
Msh5	to be determined

Fig. 4.5 Monogenetic defects associated with CVID. For most cases of CVID the gene defect remains unknown. TACI deficiency, BAFF-R deficiency, and mutations in *Msh5* do not cause CVID in all affected patients, suggesting that the etiology of most cases of CVID is multifactorial. Potential disease-modifying factors are yet to be identified.

Questions.

1 Like Mrs Johnson, most CVID patients with TACI mutations are heterozygous for the mutation. These patients express both mutant and normal TACI molecules on the surface of their B cells. Why would these heterozygous patients have the disease?

- 2 Mrs Johnson and a friend, Mrs Smith, both have common variable immunodeficiency (CVID). How can two individuals with the same rare deficiency have different disorders (CVID versus IgA deficiency)?
- 3 What other monogenic defects have been described in patients with CVID?

The maturation of T lymphocytes.

Without T cells, life cannot be sustained. In Case 1 we learned that an absence of B cells was compatible with a normal life as long as infusions of immunoglobulin G were maintained. When children are born without T cells, they seem normal for the first few weeks or months. Then they begin to acquire infections, often with opportunistic pathogens, and in the absence of appropriate treatment they die while still in infancy. An absence of functional T cells causes severe combined immunodeficiency (SCID). It is called severe because it is fatal, and called combined because, in humans, B cells cannot function without help from T cells, so that even if the B cells are not directly affected by the defect, both humoral and cell-mediated immunity are lost. Unlike X-linked agammaglobulinemia, which results from a monogenic defect, SCID comprises multiple disorders that can result from any one of several different genetic defects. The incidence of SCID is three times greater in males than in females, and this male:female ratio of 3:1 is due to the fact that the most common form of SCID is X-linked. In the United States, about 25% of individuals with SCID have the X-linked form of the disease.

T-cell precursors migrate to the thymus to mature, at first from the yolk sac of the embryo, and subsequently from the fetal liver and bone marrow (Fig. 5.1). The rudimentary thymus is an epithelial anlage derived from the third and fourth pharyngeal pouches. By 8 weeks of human gestation, the invasion of precursor T cells, and of dendritic cells and macrophages, has transformed the gland into a central

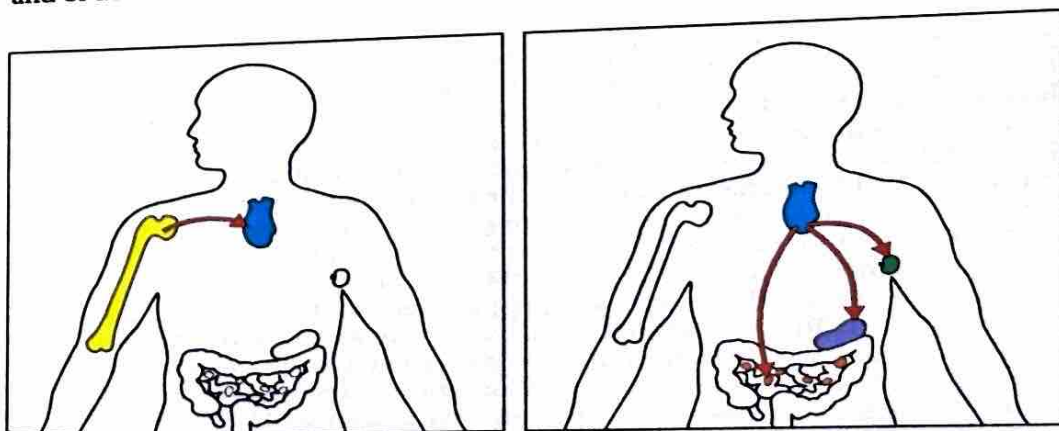


Fig. 5.1 T-cell precursors migrate to the thymus to mature. T cells derive from bone marrow stem cells, whose progeny migrate from the bone marrow to the thymus (left panel), where the development of the T cell takes place. Mature T cells leave the thymus and recirculate from the bloodstream through peripheral lymphoid tissues (right panel) such as lymph nodes, spleen, or Peyer's patches, where they may encounter antigen.

TOPICS BEARING ON THIS CASE:

The development of T cells in the thymus

Signal transduction through the T-cell receptor

Interleukin-2 receptor

Defects in T-cell function result in severe combined immunodeficiency

Testing T-cell responses with polyclonal mitogens